Three tiers of indeterminacy: the exact hull at small places and the container of Step (v) of IUT IV

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Abstract

In Step (v) of the proof of inter-universal Teichmüller theory (IUT) IV, Theorem 1.10, the holomorphic hull of the union of the possible images of the Θ-pilot object is bounded by a container that contains the region $p^{\lambda} \cdot (R_I)^{\sim}$; the indeterminacies (Ind1) and (Ind2) are accounted for by the arbitrary automorphisms ϕ of IUT IV, Proposition 1.2, and (Ind3) by taking upper bounds. At places that are small in the sense of Proposition 1.2(iii)—p odd and every ramification index at most $p-2$—we compare this container with the hulls reached by four classes of maps: unit scalars and field automorphisms; the maps induced factor by factor by the actual (Ind1) and (Ind2); all factorwise automorphisms of the log-shells; and all automorphisms ϕ of Proposition 1.2.

For the last two classes the hull is exactly $p^{\lceil \lambda \rceil - \mathfrak{d}_I} \cdot (R_I)^{\sim}$, where $\mathfrak{d}_I$ is the sum of the orders of the differents, whereas the container of Step (v) has hull $p^{\lfloor \lambda \rfloor - \mathfrak{d}_I} \cdot (R_I)^{\sim}$. So when λ is not an integer—which is always the case at bad places—the container is larger than anything these automorphisms reach by exactly a factor p, that is, by $\log p$ in log-volume. The difference comes from the proof of Proposition 1.2(ii), which uses Proposition 1.1 with the different of every factor; Proposition 1.1 applied to the factor that carries p^{λ} removes it. Unit scalars and field automorphisms do not enlarge the hull. The actual (Ind1), (Ind2) reach the exact value when λ is an integer, provided that automorphisms of the local Galois groups exist that carry a unit to the boundary of the log-shell; at small places we reduce this to one condition on the transvections found by Hoshi–Nishio and Kondo. The difference is a term of conductor type and does not change the shape of the inequality of IUT IV.

No claim is made about the truth of Corollary 3.12 of IUT III or of the *abc* conjecture.

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1 Introduction

1.1 The question

S In Step (v) of the proof of [IUT4, Thm. 1.10] (p. 27), Mochizuki writes:

> "Thus, the *inclusion* '$\phi(p^{\lambda} \cdot (R_I)^{\sim}) \subseteq p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor} \cdot \log_p(R_I^{\times})$' of Proposition 1.4, (iii), implies that the result of multiplying '$p^{\lfloor \lambda \rfloor - |I|} \cdot 2^{-|I|} \cdot \log_p(R_I^{\times})$' by a suitable *nonpositive* [...] integer power of $p_{v_{\mathbb{Q}}}$ *contains* the 'union of possible images of a Θ-pilot object' discussed in Step (iv). That is to say, the *indeterminacies* (Ind1) and (Ind2) are taken into account by the *arbitrary* nature of the automorphism 'ϕ' [cf. Proposition 1.2], while the *indeterminacy* (Ind3) is taken into account by the fact that we are considering *upper bounds* [...]"

So IUT IV bounds the effect of (Ind1) and (Ind2) by all automorphisms that preserve the lattice of logarithms of units. This note asks two questions at small places. How far do these automorphisms actually reach, and how does the container compare with that? And how much of it can the maps induced by the actual (Ind1) and (Ind2) use?

1.2 Results

(1) P For the class $\mathfrak{L}$ of all automorphisms ϕ of Proposition 1.2, and for the class $\mathfrak{L}_{\mathrm{f}}$ of factorwise automorphisms of the log-shells, the hull of the union of the images of $p^{\lambda}\cdot(R_I)^{\sim}$ is exactly $p^{\lceil\lambda\rceil-\mathfrak{d}_I}\cdot(R_I)^{\sim}$ (Theorem 3.2).

(2) P The container of Step (v) has hull $p^{\lfloor\lambda\rfloor-\mathfrak{d}_I}\cdot(R_I)^{\sim}$. When $\lambda\notin\mathbb{Z}$—always the case at bad places—it is larger by exactly a factor p; when $\lambda\in\mathbb{Z}$ the two agree (Corollary 3.3). The factor comes from the different of the factor that carries p^{λ} (Remark 3.4).

(3) P Unit scalars and field automorphisms do not enlarge the hull (Theorem 4.1). The hull for the actual (Ind1), (Ind2) is contained in the exact value of (1) (Theorem 4.2); when $\lambda\in\mathbb{Z}$ it is equal to it, provided that at each place an automorphism of the local Galois group carries some integral element to the boundary of the log-shell (Theorem 4.4). At small places this condition follows from one condition on the transvections of Hoshi–Nishio and Kondo (Proposition 4.5); that condition is not proved here.

(4) P A positive room can occur only at $p=2$ or at primes with a ramified place above them, and only through automorphisms of the local Galois groups that are not unit scalars times field automorphisms (Proposition 4.6).

(5) N For the explicit initial Θ-datum D109 of [Cat], at its bad place above 109, the container exceeds the exact value by $\log 109$ for every label (Section 5).

1.3 Relation to the author's earlier notes

P The author's note [W311] treats a global, logical question: even granting the log-volume content of all the results cited in the proof of Corollary 3.12, the conclusion does not follow. Its Theorem 22 builds a model on an actual local field, replaces the groups of (Ind1), (Ind2) by all automorphisms of the log-shells (a choice it marks as unproved), reads the possible images maximally, takes a container larger than that of IUT IV, and uses only that the enlargement of the hull is bounded by a constant E_L depending on the local field. The present note is local and numerical, and does not touch Step (xi). It determines the exact value of that enlargement at small places, compares it with the container of IUT IV, and asks how much of the replaced group the actual (Ind1), (Ind2) reach. Its values are smaller than the bound of [W311], so the conclusion of [W311] is unaffected.

The author's note [Gap] measures the (Ind2)-terms of lana-agents/iut from the region $\theta\cdot\bigotimes\mathcal{I}$ built on the log-shells; there the group of all factorwise automorphisms and the class of Proposition 1.2 give the same value, and the span is the container of Step (v) (Remark 3.5). The third report [Third] computes the floor slack $1+\lfloor\lambda\rfloor-\lambda$ of Proposition 1.4(iii), and the fourth report [Fourth] records that $\mathrm{Ism}(G_K)=\mathbb{Z}_p^{\times}$ for a finite extension K of $\mathbb{Q}_p$, after Minamide–Tsujimura [MT].

1.4 Relation to earlier work

Section 7 gives details and quotations. In short: Proposition 1.1 of IUT IV is stated with a free choice of the factor that does not pay its different, but the proof of Proposition 1.2(ii) uses all factors; Dupuy–Hilado note that one may choose the factor that makes the inclusion tightest, and exempt the factor with the largest different [DHPS, Rem. 2.8.2, (6.4)]; lana-agents/iut

states that for $\lambda = 0$ the hull is exactly the container of IUT IV, and that the bound of IUT IV adds 1 for the rounding [lana]. Mochizuki states that automorphisms of the log-shells give only bounded discrepancies [EssLgc, Ex. 3.5.1], and that the valuation is the data "deformed" by automorphisms that do not arise from field automorphisms [AbsTopIII, §I3]. The exact value with the ceiling for the region $p^{\lambda} \cdot (R_I)^{\sim}$, and the difference of exactly $\log p$ from the container when $\lambda \notin \mathbb{Z}$, were not found in the literature we searched. The transvection formula of Section 4.1 follows in substance from Hoshi–Nishio [HN] and Kondo [Kondo]; it is not new.

1.5 What is not claimed

Nothing about the truth of Corollary 3.12 or of the *abc* conjecture. That the family of possible images considered here is the family of IUT: it does not include (Ind3), it takes the Kummer isomorphisms to be the identity, and it treats label permutations as IUT IV does (Section 2.4). How much room (Ind3) provides. Anything at places that are not small. The effect on the final constants of Theorem 1.10; we do not compute it.

1.6 Labels

S a statement about a cited text; X a statement of another author; P proved here; N checked by an exact computation (scripts in the deposit); H not proved.

2 Setting

2.1 Step (v)

S We use the notation of [IUT4, Prop. 1.1, 1.2, 1.4, Thm. 1.10, Steps (iv), (v)]. Let p be an odd prime, $j \in \{1, \dots, \ell^*\}$ a label and $I = \mathbb{S}^{\pm}_{j+1} = \{0, 1, \dots, j\}$. For each $i \in I$ let v_i be a place of K above p (not necessarily distinct), $k_i = K_{v_i}$, $R_i = \mathcal{O}_{k_i}$, e_i its ramification index, π_i a uniformizer, and $\mathfrak{d}_i$ the order of a generator of the different of $R_i/\mathbb{Z}_p$, with $\mathrm{ord}(p) = 1$. Put $R_I = \bigotimes_{\mathbb{Z}_p} R_i$, let $(R_I)^{\sim}$ be its normalization, $T = R_I \otimes \mathbb{Q}_p$, and $\mathfrak{d}_I = \sum_i \mathfrak{d}_i$. Following Step (v), put $i^{\dagger} = j$, let $\lambda = \mathrm{ord}(\underline{q}^{j^2}_{=v_j})$ if v_j is bad and $\lambda = 0$ if it is good, so that $\lambda \in \frac{1}{e_j}\mathbb{Z}$, and let $p^{\lambda} \in k_j$ have order λ. The region is

$$\Theta = p^{\lambda} \cdot (R_I)^{\sim} \qquad (p^{\lambda} \text{ in the factor } j).$$

For $x \in T$ written as $p^k \prod_i \pi_i^{a_i}$ $(k, a_i \in \mathbb{Z})$ the order $\mathrm{ord}(x) = k + \sum_i a_i/e_i$ is the same in every component of T, and $\mu^{\log}(x \cdot (R_I)^{\sim}) - \mu^{\log}((R_I)^{\sim}) = -\mathrm{ord}(x) \log p$ for the normalized log-volume of [IUT4, Prop. 1.4].

2.2 Small places

Definition 2.1. A place $v \mid p$ is *small* if p is odd and $e_v \le p - 2$ (as in [Gap]); p is small if every place above it is small.

Lemma 2.2. P *If all factors are small, then:*

(a) each $k_i/\mathbb{Q}_p$ is tamely ramified, $\mathfrak{d}_i = (e_i - 1)/e_i$, $a_i = 1/e_i$, hence $\mathfrak{d}_i + a_i = 1$;

(b) the log-shell is $\mathcal{I}_i = (2p)^{-1} \log_p(R_i^{\times}) = \pi_i^{1-e_i} R_i$;

(c) the module of the inclusion used in Step (v) is $p^{\lfloor \lambda - \mathfrak{d}_I - a_I \rfloor} \cdot \log_p(R_I^{\times}) = p^{\lfloor \lambda \rfloor - |I|} \cdot 2^{-|I|} \cdot \log_p(R_I^{\times}) = p^{\lfloor \lambda \rfloor} \cdot \bigotimes_i \mathcal{I}_i$, so the "suitable nonpositive integer power" can be taken to be p^0. We call it the container *and write C.*

Proof. (a) $e_i \le p-2 < p$, so $p \nmid e_i$ and the different is $\pi_i^{e_i-1}R_i$; $\lceil e_i/(p-2)\rceil = 1$. (b) S [IUT4, Prop. 1.2(i)]: for $p>2$ and $e_i \le p-2$, $\log_p(R_i^\times) = p^{a_i}\cdot R_i = \pi_i R_i$; since p is odd, 2 is a unit and $(2p)^{-1}\pi_i R_i = \pi_i^{1-e_i}R_i$. (c) By (a), $\mathfrak{d}_I + a_I = |I|$ is an integer. □

2.3 Hull and room

S The holomorphic hull $\mathrm{hull}(X)$ of $X \subseteq T$ is the intersection of the sets $c\cdot(R_I)^\sim$, $c \in T^\times$, that contain X [IUT3, Rem. 3.9.5(i)]. For a set $\mathfrak{X}$ of $\mathbb{Q}_p$-linear automorphisms of T let $U_{\mathfrak{X}} = \bigcup_{g\in\mathfrak{X}} g(\Theta)$ and define the *room*

$$\rho(\mathfrak{X}) = \mu^{\log}(\mathrm{hull}(U_{\mathfrak{X}})) - \mu^{\log}(\Theta);$$

for a module $M \supseteq \Theta$ write $\rho(M) = \mu^{\log}(\mathrm{hull}(M)) - \mu^{\log}(\Theta)$.

2.4 Four classes

S In [IUT3, Thm. 3.11(i)] (p. 154), (Ind1) consists of "the indeterminacies induced by the automorphisms of the procession of $\mathcal{D}^\vdash$-prime-strips", and (Ind2), at a nonarchimedean $v_\mathbb{Q}$, of "the indeterminacies induced by the action of independent copies of Ism [...] on each of the direct summands of the $j+1$ factors appearing in the tensor product". An automorphism of a $\mathcal{D}^\vdash$-prime-strip is, at each place, an automorphism of the topological group G_v, which need not arise from a field automorphism.

Definition 2.3. Consider the following sets of $\mathbb{Q}_p$-linear automorphisms of T:

- $\mathfrak{N}$: the maps $\bigotimes_i c_i\sigma_i$ with $c_i \in \mathbb{Z}_p^\times$ and $\sigma_i \in \mathrm{Aut}(k_i/\mathbb{Q}_p)$;
- $\mathfrak{A}$: the maps $\bigotimes_i c_i\gamma_i$ with $c_i \in \mathbb{Z}_p^\times$ and γ_i the map induced on k_i, through the mono-analytic reconstruction of the log-shell, by an automorphism β_i of the topological group G_{k_i};
- $\mathfrak{L}_\mathrm{f}$: the maps $\bigotimes_i g_i$ with g_i a $\mathbb{Q}_p$-linear automorphism of k_i such that $g_i(\mathcal{I}_i) = \mathcal{I}_i$;
- $\mathfrak{L}$: the $\mathbb{Q}_p$-linear automorphisms ϕ of T with $\phi(\bigotimes_i \mathcal{I}_i) = \bigotimes_i \mathcal{I}_i$, that is, the automorphisms of Proposition 1.2 of [IUT4].

Proposition 2.4. P $\mathfrak{N} \subseteq \mathfrak{A} \subseteq \mathfrak{L}_\mathrm{f} \subseteq \mathfrak{L}$.

Proof. A field automorphism of k_i over $\mathbb{Q}_p$ is induced by conjugation by an extension to an algebraic closure, and [Fourth, Prop. 9] applies [MT, Thm. F, Thm. 7.4]: $\mathrm{Ism}(G_{k_i}) = \mathbb{Z}_p^\times$, acting by unit scalars. The mono-analytic log-shell is constructed from the $\mathcal{D}^\vdash$-prime-strip by a functorial algorithm [IUT3, Prop. 1.2(vi)], so the induced maps preserve the log-shells. Tensor products of lattice automorphisms preserve the tensor product of the lattices. □

Remark 2.5 (label permutations and (Ind3)). (Ind1) also permutes the labels. As in Step (v), we fix the factor $i^\dagger = j$ carrying the theta value; IUT IV symmetrizes over the choice of $i^\dagger$ when it averages [IUT4, p. 28], and lana-agents/iut likewise takes the hull for each (Ind1)-image separately [lana]. We do not include (Ind3).

3 The exact hull for the classes of Proposition 1.2

Proposition 3.1. P *If all factors are small, then* $\Theta \subseteq p^{\lceil\lambda\rceil}\cdot\bigotimes_i \mathcal{I}_i$.

Proof. S [IUT4, Prop. 1.1]: "Fix an element $*\in I$; write $I^* = I\setminus\{*\}$. Then $p^{\mathfrak{d}_{I^*}}\cdot(R_I)^\sim \subseteq R_I \subseteq (R_I)^\sim$," where $p^{\mathfrak{d}_{I^*}}$ is a product of generators $\delta_i \in R_i$ of the differents for $i \in I^*$. Take $* = j$. Then $\Theta \subseteq p^\lambda R_j \otimes \bigotimes_{i\neq j}\delta_i^{-1}R_i$. By Lemma 2.2, $\delta_i^{-1}R_i = \pi_i^{1-e_i}R_i = \mathcal{I}_i$. Finally

$p^\lambda R_j \subseteq p^k \mathcal{I}_j = \pi_j^{e_j k+1-e_j} R_j$ if and only if $k \le \lambda + (e_j-1)/e_j$; as $\lambda \in \frac{1}{e_j}\mathbb{Z}$, the largest such k is $\lceil \lambda \rceil$. □

Theorem 3.2 (exact hull). P *Suppose all factors are small, and put* $\mu = p^{\lceil \lambda \rceil} \prod_i \pi_i^{1-e_i} \in T^\times$. *For* $\mathfrak{X} = \mathfrak{L}_\mathrm{f}$ *and for* $\mathfrak{X} = \mathfrak{L}$,

$$\mathrm{hull}(U_\mathfrak{X}) = \mu \cdot (R_I)^\sim = p^{\lceil \lambda \rceil - \mathfrak{d}_I} \cdot (R_I)^\sim, \qquad \rho(\mathfrak{X}) = \big(\mathfrak{d}_I - (\lceil \lambda \rceil - \lambda)\big) \log p.$$

Proof. Upper bound: every $\phi \in \mathfrak{L}$ preserves $p^{\lceil \lambda \rceil} \bigotimes_i \mathcal{I}_i = \mu R_I$, so $U_\mathfrak{L} \subseteq \mu R_I$ by Proposition 3.1. Since $1 \in R_I \subseteq (R_I)^\sim$, a set $c \cdot (R_I)^\sim$ containing μR_I contains μ, hence $\mu \cdot (R_I)^\sim$; so $\mathrm{hull}(\mu R_I) = \mu \cdot (R_I)^\sim$ and $\mathrm{hull}(U_\mathfrak{L}) \subseteq \mu \cdot (R_I)^\sim$.

Lower bound: the automorphism group of a free $\mathbb{Z}_p$-module acts transitively on its primitive vectors. For $i \ne j$, both 1 and $\pi_i^{1-e_i}$ are primitive in $\mathcal{I}_i$ (they lie in $\mathcal{I}_i$ and not in $p\mathcal{I}_i = \pi_i R_i$), so some $g_i \in \mathrm{GL}(\mathcal{I}_i)$ has $g_i(1) = \pi_i^{1-e_i}$. For $i = j$, both p^λ and $p^{\lceil \lambda \rceil} \pi_j^{1-e_j}$ are primitive in $p^{\lceil \lambda \rceil} \mathcal{I}_j$ (because $e_j \lambda \le e_j \lceil \lambda \rceil$), so some $g_j \in \mathrm{GL}(\mathcal{I}_j)$ has $g_j(p^\lambda) = p^{\lceil \lambda \rceil} \pi_j^{1-e_j}$. The image of $1 \otimes \cdots \otimes 1 \otimes p^\lambda \in \Theta$ under $\bigotimes_i g_i \in \mathfrak{L}_\mathrm{f}$ is $\mu \in T^\times$, so $\mathrm{hull}(U_{\mathfrak{L}_\mathrm{f}}) \supseteq \mu \cdot (R_I)^\sim$. With $U_{\mathfrak{L}_\mathrm{f}} \subseteq U_\mathfrak{L}$ both hulls equal $\mu \cdot (R_I)^\sim$.

Room: $\mathrm{ord}(\mu) = \lceil \lambda \rceil - \mathfrak{d}_I$ by Lemma 2.2(a), and $\mathrm{ord}(p^\lambda) = \lambda$. □

Corollary 3.3 (container and bound). P *Under the hypotheses of Theorem 3.2,*

$$\begin{aligned} \mathrm{hull}(C) &= p^{\lfloor \lambda \rfloor - \lceil \lambda \rceil} \cdot \mathrm{hull}(U_\mathfrak{L}), \\ \rho(C) - \rho(\mathfrak{L}) &= (\lceil \lambda \rceil - \lfloor \lambda \rfloor) \log p, \\ (\mathfrak{d}_I + 1) \log p - \rho(\mathfrak{L}) &= (1 + \lceil \lambda \rceil - \lambda) \log p. \end{aligned}$$

In particular, if $\lambda \notin \mathbb{Z}$ *the hull of the container is* p^{-1} *times the hull reached by all of* $\mathfrak{L}$ *(its log-volume is larger by exactly* $\log p$*), and if* $\lambda \in \mathbb{Z}$ *they coincide. Here* $(\mathfrak{d}_I + 1) \log p$ *is the room allowed by [IUT4, Prop. 1.2(iii)],* $\phi(p^\lambda \cdot (R_I)^\sim) \subseteq p^{\lambda - \mathfrak{d}_I - 1} \cdot (R_I)^\sim$.

Proof. By Lemma 2.2(c), $C = p^{\lfloor \lambda \rfloor} \bigotimes_i \mathcal{I}_i$, whose hull is $p^{\lfloor \lambda \rfloor} \prod_i \pi_i^{1-e_i} \cdot (R_I)^\sim$ as in the proof of Theorem 3.2. Compare orders. □

Remark 3.4 (where the factor p comes from). S The proof of [IUT4, Prop. 1.2(ii)] (p. 11) reads: "Observe that it follows from the first displayed inclusion [*of R_I-modules*!] of Proposition 1.1 that $p^{\mathfrak{d}_I + a_I} \cdot (R_I)^\sim \subseteq \bigotimes_{i \in I} p^{a_i} \cdot R_i$". P Here the different of every factor is paid. Using Proposition 1.1 with $* = i^\dagger$ instead gives, by the same argument,

$$\phi(p^\lambda \cdot (R_I)^\sim) \subseteq p^{\lfloor \lambda - \mathfrak{d}_{I \setminus \{i^\dagger\}} - a_I \rfloor} \cdot \log_p(R_I^\times).$$

If all factors are small, $\mathfrak{d}_{I \setminus \{i^\dagger\}} + a_I = |I| - 1 + 1/e_j$, and for $\lambda \in \frac{1}{e_j}\mathbb{Z}$ one has $\lfloor \lambda - 1/e_j \rfloor = \lceil \lambda \rceil - 1$, so the right-hand side is exactly $p^{\lceil \lambda \rceil} \bigotimes_i \mathcal{I}_i$, the module of Proposition 3.1. So the factor p of Corollary 3.3 is the price of paying the different of the factor that carries p^λ. The remaining $(1 + \lfloor \lambda \rfloor - \lambda) \log p$ between the container and the bound of Proposition 1.2(iii) is the floor slack of Proposition 1.4(iii), computed on explicit data in [Third, § The floor]. None of this is a defect of IUT IV: the estimate is an upper bound, and an upper bound may be generous.

Remark 3.5 (the frame of [Gap]). P [Gap] measures from the region $\theta \cdot \bigotimes \mathcal{I}$ built on the log-shells. There the factorwise group (`tensorAut` of lana-agents/iut) and the class of Proposition 1.2 give the same value, and the span of the union is $p^\kappa \bigotimes \mathcal{I}$ with $\kappa = \lfloor \mathrm{ord}_p \theta \rfloor$ at small places, which is the container C [Gap, Thm. 3.2, §6.2]. Here we measure from the region $\Theta = \theta \cdot (R_I)^\sim$ of Step (v) itself. Again the two classes agree (Theorem 3.2), but their common value is smaller than C by a factor p when $\lambda \notin \mathbb{Z}$. The two statements are compatible because the starting regions differ: $\rho(\theta \cdot \bigotimes \mathcal{I}) = \mathfrak{d}_I \log p$ exceeds $\rho(\mathfrak{L}) = (\mathfrak{d}_I - (\lceil \lambda \rceil - \lambda)) \log p$.

Remark 3.6 (checks). N For $p = 5$ and all factors equal to $k = \mathbb{Q}_5(\sqrt{5})$ ($e = 2 \le p - 2$), with two factors ($j = 1$) and three factors ($j = 2$), we computed by exact linear algebra the largest k with $\Theta \subseteq 5^k \bigotimes \mathcal{I}$, for $\theta = \sqrt{5}^m$ in the last factor: it is $\lceil m/2 \rceil$ for $0 \le m \le 12$ (two factors) and $0 \le m \le 7$ (three factors). For random elements of $\mathrm{GL}(\mathcal{I})$ in each factor (400 and 150 samples per m for two factors, 150 for three) the componentwise minimal valuation of the images is $2\lceil m/2 \rceil - (j+1)$ in $\sqrt{5}$-units, the value of Theorem 3.2; the container gives $2\lfloor m/2 \rfloor - (j+1)$, which differs by 2 exactly when m is odd. For three factors and $m = 0$ the hull has radius $5^{3/2}$, in agreement with the statement of lana-agents/iut quoted in Section 7.

4 The actual indeterminacies

Theorem 4.1. P $\rho(\mathfrak{N}) = 0$.

Proof. $\sigma_j(p^\lambda)$ is a unit multiple of p^λ, and $c_i \sigma_i$ preserves R_i, hence the normalization; so every element of $\mathfrak{N}$ maps Θ onto itself. □

Theorem 4.2. P *If all factors are small, then* $\mathrm{hull}(U_{\mathfrak{A}}) \subseteq p^{\lceil\lambda\rceil - \mathfrak{d}_I} \cdot (R_I)^\sim$, *and* $\rho(\mathfrak{A}) \le (\mathfrak{d}_I - (\lceil\lambda\rceil - \lambda)) \log p$.

Proof. Proposition 2.4 and Theorem 3.2. □

Definition 4.3. Say that a place v satisfies *(U)* if there are an automorphism β of G_{K_v} and $x \in \mathcal{O}_{K_v}$ such that the induced map γ satisfies $|\gamma(x)| = |\pi_v|^{1-e_v}$, that is, γ carries an integral element to the boundary of the log-shell. (For $e_v = 1$ the identity does.)

Theorem 4.4. P *Suppose all factors are small,* $\lambda \in \mathbb{Z}$, *and every* v_i *satisfies (U). Then* $\mathrm{hull}(U_{\mathfrak{A}}) = p^{\lambda - \mathfrak{d}_I} \cdot (R_I)^\sim$, *the value of Theorem 3.2.*

Proof. Take x_i, γ_i as in (U) for v_i. Since x_i is integral and $p^\lambda \in \mathbb{Q}_p$, $y = x_0 \otimes \cdots \otimes x_{j-1} \otimes p^\lambda x_j \in \Theta$. Its image under $\bigotimes_i \gamma_i \in \mathfrak{A}$ is $p^\lambda \gamma_0(x_0) \otimes \cdots \otimes \gamma_j(x_j)$, a pure tensor of nonzero elements, hence a unit of T of order $\lambda + \sum_i (1 - e_i)/e_i = \lambda - \mathfrak{d}_I$ in every component. So the hull contains $p^{\lambda - \mathfrak{d}_I} \cdot (R_I)^\sim$; Theorem 4.2 gives the other inclusion. □

4.1 When does (U) hold?

X Automorphisms of G_L that do not preserve the ring of integers of $\overline{L}$ exist [Hoshi, Rem. 4.3.1(ii)], and the group-theoretic log-shell equals the ring of integers exactly when p is odd and k is unramified, or $k \cong \mathbb{Q}_2$ or $\mathbb{Q}_3(\zeta_3)$ [HoshiTop, Thm. A(ii)]. Every such automorphism α induces a map α_+ that is compatible with $\mathrm{Tr}_{k/\mathbb{Q}_p}$ and restricts to an automorphism of its kernel [HN, Lemma 2.3(ii)]. For p odd and $[k : \mathbb{Q}_p]$ even, the Jannsen–Wingberg presentation [JW] as made explicit by Kondo gives an automorphism ϕ (with $\phi(x_2) = x_2 x_1$) whose induced map moves one generator of the logarithms of units by a multiple of another, while the kernel of the trace is spanned by the others [Kondo, Thm. 1.3, Lemma 1.5]; Hoshi–Nishio give unipotent induced maps α_+ with $(\alpha_+^n - \mathrm{id})^2 = 0$ [HN, Thm. 1.5]. Together these give induced maps of the form

$$x \longmapsto x + \mathrm{Tr}_{k/\mathbb{Q}_p}(x)\,\eta, \qquad \mathrm{Tr}_{k/\mathbb{Q}_p}(\eta) = 0.$$

This formula follows in substance from [HN, Kondo]; we do not claim it.

Proposition 4.5. P *Let* v *be small with* $e_v \ge 2$. *If some automorphism of* G_{K_v} *induces* $x \mapsto x + \mathrm{Tr}(x)\eta$ *with* $|\eta| = |\pi_v|^{1-e_v}$, *then* v *satisfies (U).*

Proof. $K_v/\mathbb{Q}_p$ is tame, so $\mathrm{Tr}(\mathcal{O}_{K_v}) = \mathbb{Z}_p$; take $x \in \mathcal{O}_{K_v}$ with $\mathrm{Tr}(x) = 1$. Then $\gamma(x) = x + \eta$ with $|\eta| > 1 \ge |x|$, so $|\gamma(x)| = |\eta|$. □

H Since the induced maps preserve the log-shell, $|\eta| \le |\pi_v|^{1-e_v}$ always; whether equality holds for a suitable presentation is not decided here. It concerns the position of the units attached to the generators, and does not follow from the statements we cite. For $[K_v : \mathbb{Q}_p]$ odd, and for $\lambda \notin \mathbb{Z}$ (which needs a move by $e_j\lambda \bmod e_j - 1$ steps in the last factor rather than $e_j - 1$), we have nothing.

4.2 Where the room can come from

Proposition 4.6. P *(a) If p is odd and every place above p is unramified, then $\rho(\mathfrak{L}) = 0$; so positive room needs $p = 2$ or a ramified place above p. (b) If $\rho(\mathfrak{A}) > 0$, then some element of $\mathfrak{A}$ has a factor $c_i\gamma_i$ with γ_i not of the form $c\,\sigma$ ($c \in \mathbb{Z}_p^\times$, σ a field automorphism).*

Proof. (a) S [IUT4, Prop. 1.2(iv)]: if $p > 2$ and $e_i = 1$ for all i, then $\phi((R_I)^\sim) \subseteq (R_I)^\sim$. Here $\lambda \in \mathbb{Z}$, so p^λ commutes with ϕ. (b) Theorem 4.1. □

Remark 4.7 (the inequality of Step (xi)). P Write the inequality of the form of Step (xi-f) for the family generated by $\mathfrak{A}$ (without (Ind3)) with the normalizations of IUT IV: the q-pilot has log-volume $-|\log q|$ and the native Θ-pilot, averaged over the labels, $-c\,|\log q|$ with $c = \ell(\ell+1)/12$. The inequality then reads $(c-1)|\log q| \le A + \sum_p \bar\rho_p$, where A is the archimedean part and $\bar\rho_p$ the room at p averaged over the labels. When the height is so large that $A < (c-1)|\log q|$, some $\bar\rho_p$ must be positive, so by Proposition 4.6 some actual automorphism of a local Galois group that is not a unit scalar times a field automorphism must move the region, at $p = 2$ or at a ramified prime. This is a necessary condition only; it says nothing about whether the inequality holds. At small primes $\bar\rho_p$ is at most the label average of the exact value of Theorem 3.2. Consistently with [EssLgc, p. 101], none of these terms involves the size of the q-orders.

5 An example

N The explicit initial Θ-datum D109 of [Cat] ($\ell = 7$, $\ell^* = 3$) has a bad place above $p = 109$ with $e = 105 \le 107$ and $\mathrm{ord}_p(\underline{q}) = 29/7$; the other place above 109 is good, with $e = 1$, so 109 is small. Take the component with every factor at this bad place. Then $\lambda = 29j^2/7 \notin \mathbb{Z}$ for $j = 1, 2, 3$ (as $7 \nmid 29$) and $\mathfrak{d}_I = 104(j+1)/105$. In units of $\log 109$:

j	λ	$\mathfrak{d}_I$	exact (Thm. 3.2)	container	IV bound	container − exact
1	29/7	208/105	118/105	223/105	313/105	1
2	116/7	104/35	89/35	124/35	139/35	1
3	261/7	416/105	341/105	446/105	521/105	1

Here "IV bound" is the room $(\mathfrak{d}_I + 1)\log p$ allowed by Proposition 1.2(iii). With $\log 109 \approx 4.691$ the three exact rooms are about 5.272, 11.929 and 15.236. Which bad places are selected for $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ does not affect this local computation.

6 Discussion

- P The estimate of IUT IV is an upper bound, and nothing in its logic is affected by the container being larger than what is reached. Corollary 3.3 states precisely how generous it is at small places. The difference $(\lceil\lambda\rceil - \lfloor\lambda\rfloor)\log p$ does not involve the size of the q-orders: it is a term of conductor type and does not change the shape of the height inequality.

- P At small places, enlarging the factorwise automorphisms $\mathfrak{L}_{\mathrm{f}}$—the group `tensorAut` of lana-agents/iut—to all of $\mathfrak{L}$ does not change the hull. Any effect of the enlargement must come from places that are not small.
- P The difference between the actual $\mathfrak{A}$ and $\mathfrak{L}_{\mathrm{f}}$ is a question about automorphisms of absolute Galois groups that do not come from field automorphisms: how far they move integral structures. Unit scalars and field automorphisms give no room (Theorem 4.1).
- H We do not include (Ind3). IUT IV accounts for it by considering upper bounds, and Mochizuki states that (Ind3) by itself accounts for the possibility of confusing q with q^{j^2} [EssLgc, p. 101]. Where the family of possible images with (Ind3) lies between the exact value and the container is not decided here.
- P The bound E_L of [W311, Thm. 22(a)] reads, at small places where $b_I + a_I = 0$, $\frac{1}{3}\sum_j(\mathfrak{d}_{I_j}+1+c_j)\log p$; that note reads the possible images maximally and takes a larger container, so its bound exceeds the label average of the exact value of Theorem 3.2 by $\frac{1}{3}\sum_j(1+c_j+\lceil\lambda_j\rceil-\lambda_j)\log p$. Its argument uses only the upper bound, so this does not affect its conclusion.

7 Relation to earlier work

- S The use of Proposition 1.1. [IUT4, Prop. 1.1] is stated with a free choice of $*$, while the proof of Proposition 1.2(ii) uses $p^{\mathfrak{d}_I+a_I}$ (Remark 3.4). Yamashita restates the $*$-form in his Lemma 1.4 and uses $\lceil\mathfrak{d}_I\rceil$ in Lemma 1.6, obtaining $(-\lambda+\mathfrak{d}_I+4)\log p$ [Yam, pp. 13–19]. X Dupuy and Hilado write "We can chose some K_i which makes the inclusion tightest" [DHPS, Rem. 2.8.2, p. 10], and in step (6.4) exempt the factor with the largest different [DHPS, p. 20]. Mochizuki–Fesenko–Hoshi–Minamide–Porowski change the numerical constants of Theorem 1.10 but not Propositions 1.1–1.4 [MFHMP, proof of Thm. 5.1].
- X The case $\lambda = 0$. lana-agents/iut states (file `Iut/WallEquiv/VcI.md`, first committed 13 October 2026): "There the hull is IUT IV's container $p^{-d_I}\cdot(R_I)^{\sim}$ exactly (IUT IV, Prop. 1.4(iii) with $\lambda = 0$, $I^* = \emptyset$; IUT's bound $(d_I+1)\cdot\log p$ adds 1 for the rounding $\lfloor\lambda - d_I - a_I\rfloor$)." [lana]. The author's [Gap] computes the span of the orbit for the log-shell-based region (Remark 3.5), using the same lattice lemma as the lower bound of Theorem 3.2; that lemma is elementary and not new.
- X Bounded room. Mochizuki writes that "automorphisms of the $\mathbb{Z}_p$-module I_k only give rise to bounded discrepancies in the orders of nonzero elements of $\mathcal{O}_k$" [EssLgc, Ex. 3.5.1], and that the confusion between q-parameters and their large powers "can never occur as a consequence of (Ind1), (Ind2), i.e., both of which amount to automorphisms of the [underlying topological module of the] log-shells involved" [EssLgc, p. 101]. In the proof of Corollary 3.12, Step (x), the resulting log-volumes "are invariant with respect to the indeterminacies (Ind1), (Ind2)" [IUT3, p. 181]; this concerns each image, not the hull of their union. Takkun-kohinata/IUT_LEAN (15 July 2026) states, in a comment of `Thm311.lean`, that (Ind1), (Ind2) preserve volume and so do not contribute to the volume estimate of the hull (our paraphrase) [Takkun]; again this concerns each image. Zeng computes the index by which the hull enlarges the integral structure of a tensor product of rings of integers [Zeng].
- X Non-geometric automorphisms. Mochizuki writes that the valuation data "is precisely the data that is 'deformed' by automorphisms of k that do not arise from field automorphisms" [AbsTopIII, §I3], describes (Ind1) as arising from automorphisms of G_k that do not in general preserve the ring structure [Alien, §3.7(i)], and in an e-mail quoted by

Joshi writes that such automorphisms "can act in a fashion that fails to preserve differents, discriminants" and other quantities [JoshiAnab, Appendix 16]. Scholze and Stix note: "Such isomorphisms exist for Galois groups of p-adic fields, but this did not seem to enter the discussion in a critical way" [SS, fn. 4]. Mochizuki's formalization report mentions "a small discrepancy (arising from (Ind1,2,3) + hull)" [Fm26, p. 11].

- X The classes. Dupuy and Hilado treat (Ind1), (Ind2) as acting through the automorphisms of the $\mathbb{Z}_p$-lattices of the log-shells [DH1, p. 17, §4.8]. lana-agents/iut uses `tensorAut`, which contains the image of Ism and is described as a deliberate enlargement [lana]. Minamide and Tsujimura write "Theorem F implies that (Ind2) is relatively small" [MT]. Joshi reads (Ind1) as anabelomorphy and (Ind2) as a choice of arithmetic holomorphic structure [JoshiIII, §8.11.2], a reading different from the one used here.
- P What is added here: the exact value with the ceiling for the region $p^{\lambda} \cdot (R_I)^{\sim}$ of Step (v); the difference of exactly $\log p$ from the container when $\lambda \notin \mathbb{Z}$, traced to the use of Proposition 1.1; the arrangement of the four classes, with the non-geometric automorphisms of G_v in the class $\mathfrak{A}$; and the reduction of condition (U) to one condition on the transvections.

8 Limitations

- Only small places. $p = 2$ and primes with $e_v > p - 2$ (for instance the bad place above 59 of the datum D211 of [Cat], with $e = 105$) are not treated.
- The family of possible images omits (Ind3), takes the Kummer isomorphisms to be the identity, and treats label permutations as Step (v) does.
- Condition (U) is not proved for $e_v \geq 2$, and nothing is said for $\lambda \notin \mathbb{Z}$ about the actual class.
- The effect on the final constants of [IUT4, Thm. 1.10] is not computed.

9 Reproducibility and formalization

The folder `three-tiers/` of the repository `https://github.com/turahigashi/iut-explicit-initial-theta` (tag `v0.25.0`), also deposited with this note, contains the scripts and their outputs: `check_example.py` and `check_nfactor.py` (the largest k and the random factorwise images of Remark 3.6), `check_perfactor.py` (the same for two factors with 400 samples), and `example_d109.py` (the table of Section 5, reading the place data of D109 from the program of [Walk]). They use exact rational arithmetic in Python.

Versions of Theorems 4.1, 4.2 and 4.4 and of Proposition 4.5 (with a common ramification index above p where the statements here allow different ones), and of the necessary condition in the last remark of Section 4, were also checked in Lean 4 with Mathlib, for a model of the possible images without (Ind3), in a development of the author that is not part of this deposit. The proofs given here do not rely on it. Theorem 3.2, Corollary 3.3 and Proposition 3.1 are proved on paper only.

Disclosure

The comparison studied here arose in the author's discussions with generative AI tools, and the author decided to make it the subject of this note. The proofs were found with generative AI tools (large language models) under the author's direction, and these tools were also used

for the computations, the formalization, the literature search and the writing. The author's own reading of the proofs is still in progress. The author takes full responsibility for the note.

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